

Instrumentation

Electromagnetic Sensors cont.

Lecturer: Dr. N. A. Sheini

Shahid Chamran University of Ahvaz

March 2026

حسگر القایی ترانسفورماتوری برای سنجش حرکت خطی (LVDT)

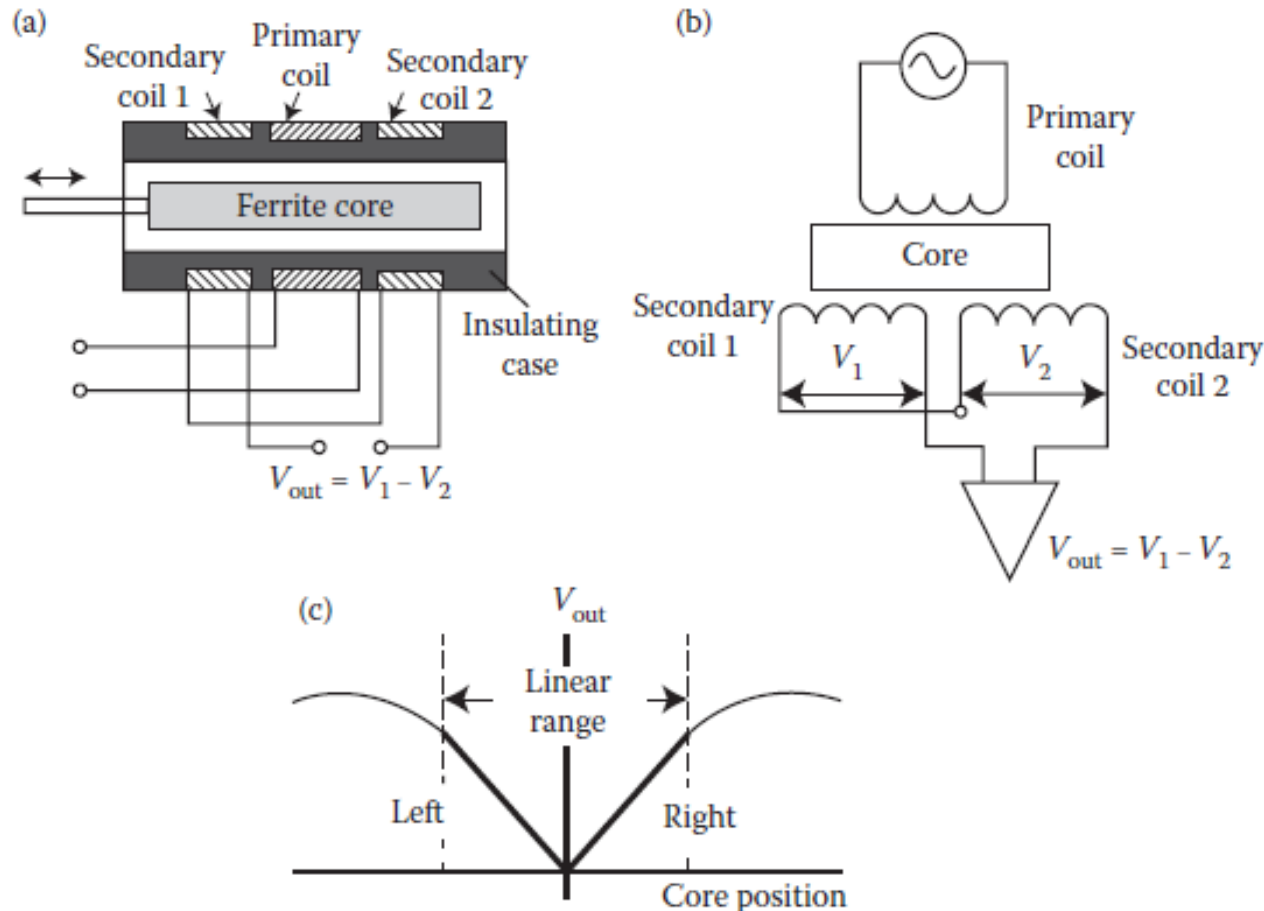


FIGURE 4.35 (a) Internal structure of an LVDT sensor, (b) its schematic diagram, and (c) its output.

حسگر القایی ترانسفورماتوری برای سنجش حرکت خطی (LVDT)

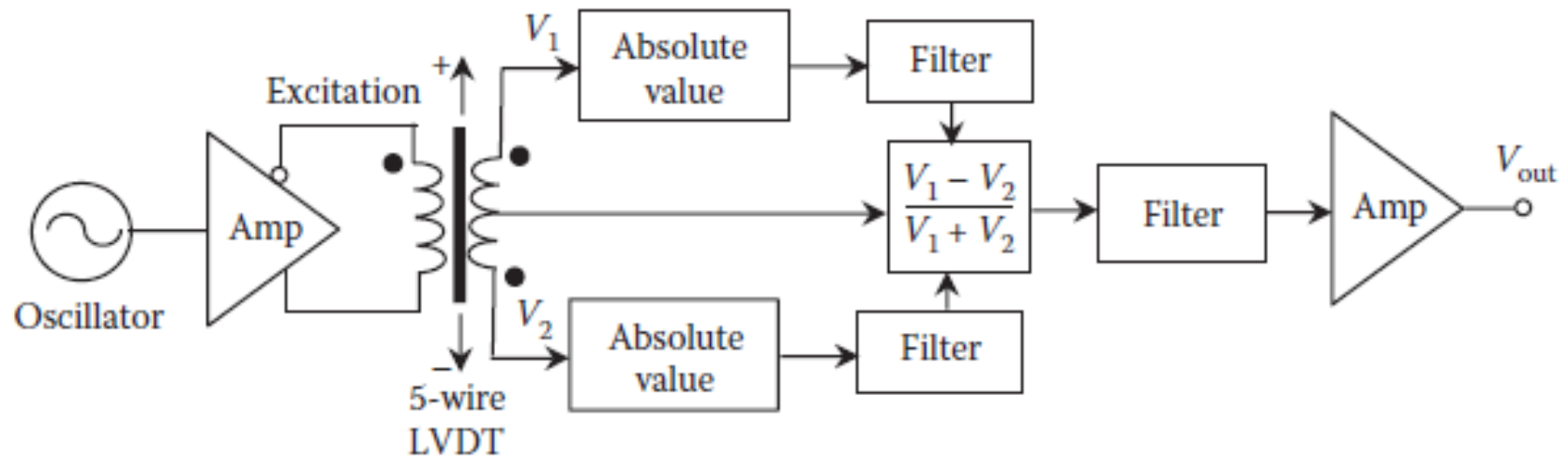
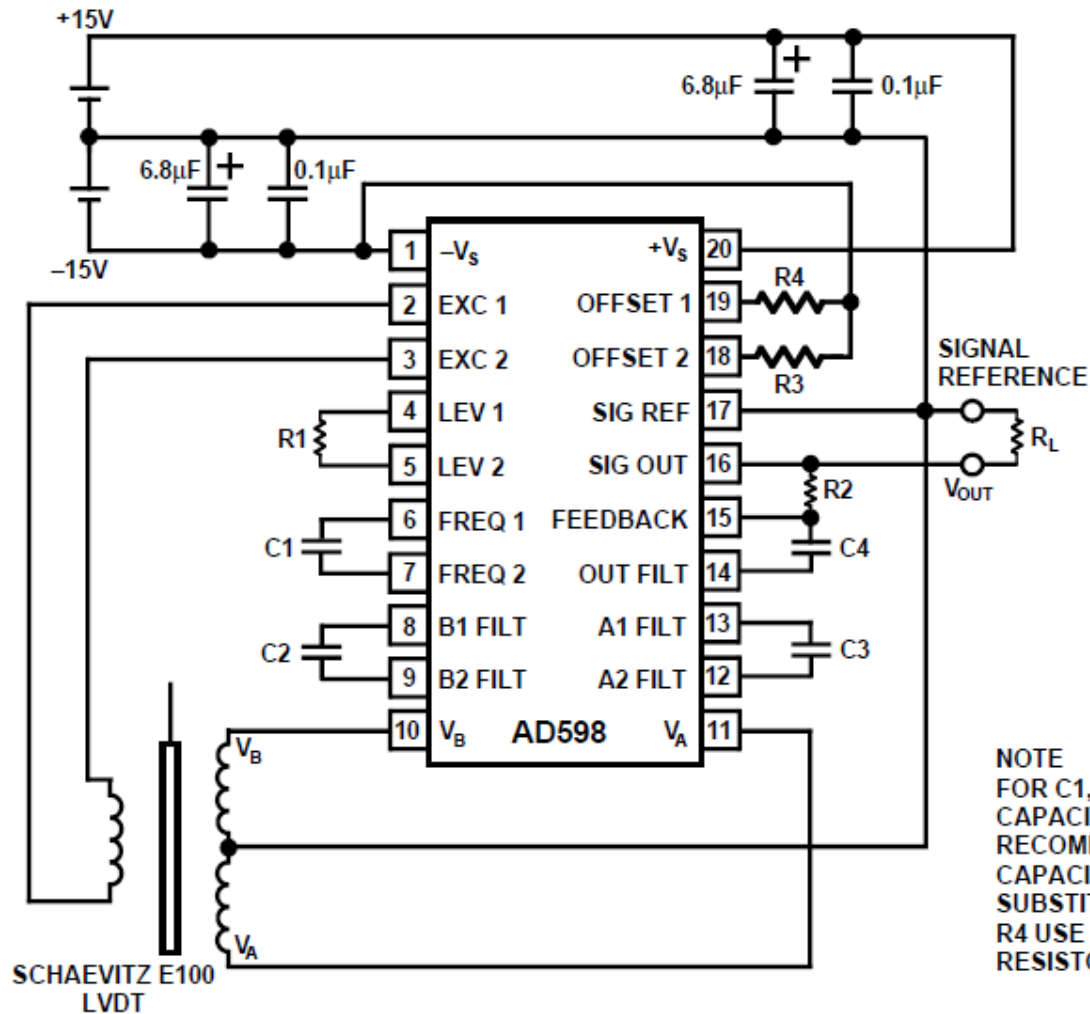


FIGURE 4.36 A simplified AD598 LVDT signal conditioner.

حسگر القایی ترانسفورماتوری برای سنجش حرکت خطی (LVDT)



حسگر القایی نوع ترانسفورماتوری – RVDT

R30A – AC Operated, Light Weight RVDT



- AC operation
- ± 60 degree angular sensing range
- Light weight
- Non-contact electrical design
- Wide operating temperature range
- Size 11 servo mount
- Anodized aluminum housing

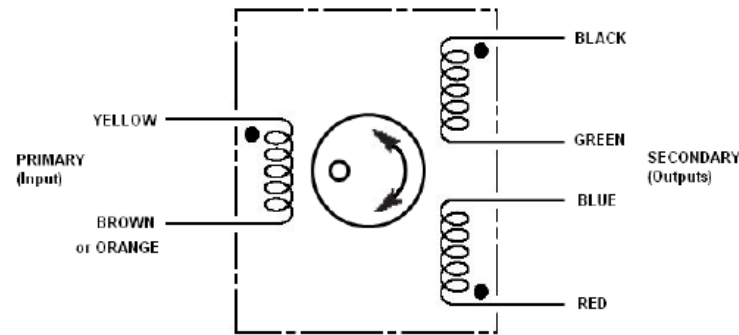
FEATURES

- High accuracy
- Infinite resolution
- Long term reliability
- Wide -55° to $+150^{\circ}\text{C}$ operating temp range
- Rugged anodized aluminum housing
- Shielded ABEC 3 precision bearings

APPLICATIONS

- Valve position
- Machine tool equipment
- Rotary actuator feedback
- Dancer arm position
- Process control

حسگر القایی نوع ترانسفورماتوری - RVDT:



Connect Green to Blue for differential output

ELECTRICAL SPECIFICATIONS						
Parameter	@10kHz Input Frequency (recommended)			@2.5kHz Input Frequency		
Angular range, degrees	±30°	±40°	±60°	±30°	±40°	±60°
Non-linearity, % of FR	±0.25%	±1%	±2%	±0.25%	±1%	±2%
Output at range ends (*)	87mV/V	116mV/V	174mV/V	69 mV/V	92 mV/V	138 mV/V
Sensitivity	2.9 mV/V/degree			2.3 mV/V/degree		
Temp coefficient of sensitivity	0.02%/°F [0.036%/°C], 20 to +160°F [-7 to +71°C]			Not specified		
Input / Output impedances	370Ω / 1300Ω			135Ω / 600Ω		
Phase shift	+3°			+35°		
Input voltage and frequency	3 VRMS @ 2.5 to 10 kHz (10kHz recommended)					
Null voltage	0.5% of FRO, maximum					

Notes:

All values are nominal unless otherwise noted

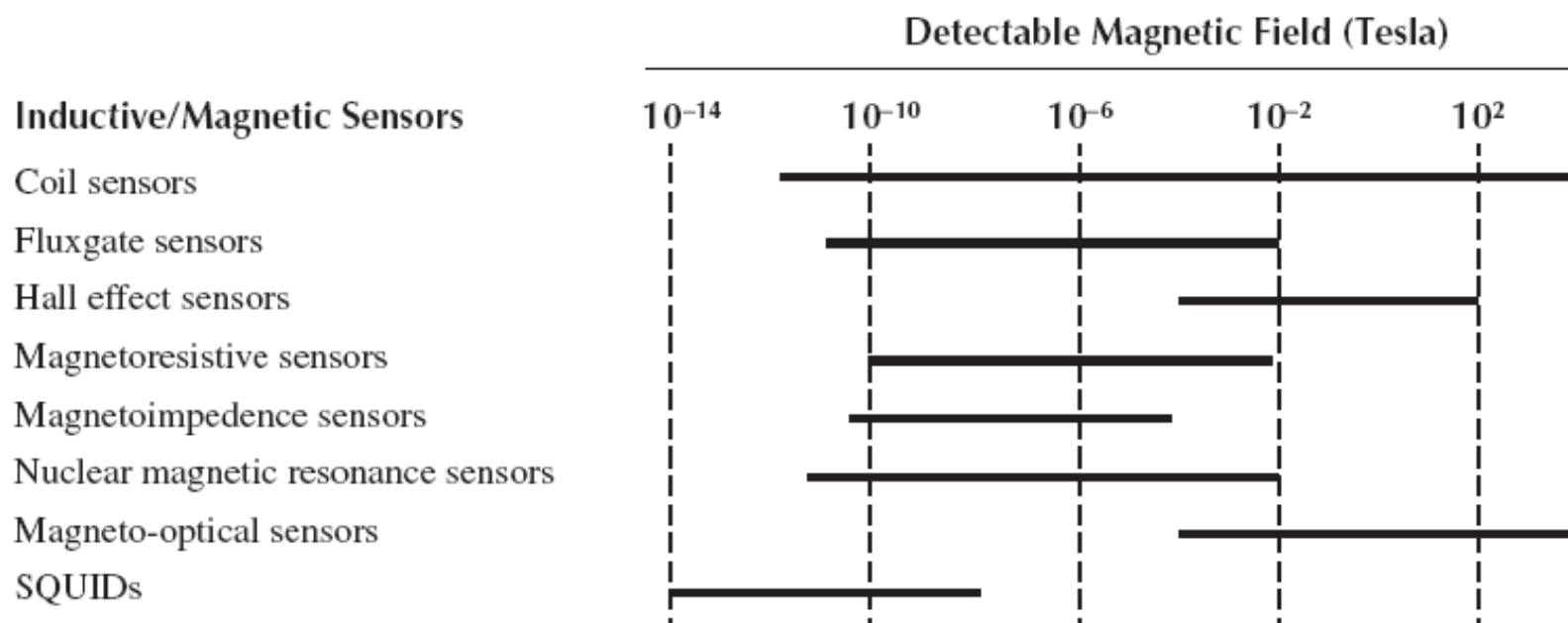
(*): Unit for output at range ends is millivolt per volt of excitation (input voltage)

FR (Full Range) is the angular range, end to end; $2 \times A^\circ$ for $\pm A^\circ$ angular range

FRO (Full Range Output): Algebraic difference in outputs measured at the ends of the range

TABLE 5.1

Magnetic Field Detection Ranges of Inductive and Magnetic Sensors



حسگرهای شارگیت (Fluxgate):

The fluxgate sensor known from 1936 (Aschennbrener et al. 1936) is still commonly used due to many advantages: resolution starting from 1pT, alternating signal modulated by low-frequency signal proportional to measured magnetic field—this way noise and offset can be relatively easy suppressed.

Operating of such sensor is based on the fact that the magnetizing curve is nonlinear (with saturation) and symmetrical. If we magnetize the ferromagnetic material periodically with magnetic field H significantly larger than “knee” of the magnetization curve, we obtain the distorted flux density (signal A). But due to symmetry of the magnetization curve, this signal contains only odd harmonics. When additional external magnetic field H_x appears, then the magnetization curve becomes asymmetrical this causes presence of even harmonics in the output signal. Usually, the second harmonic is used as the signal proportional to external magnetic field H_x .

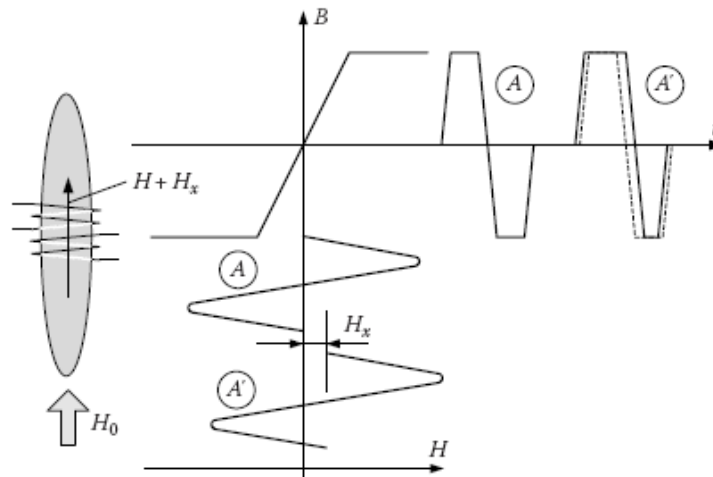


FIGURE 4.50
Principle of operation of the fluxgate sensor.

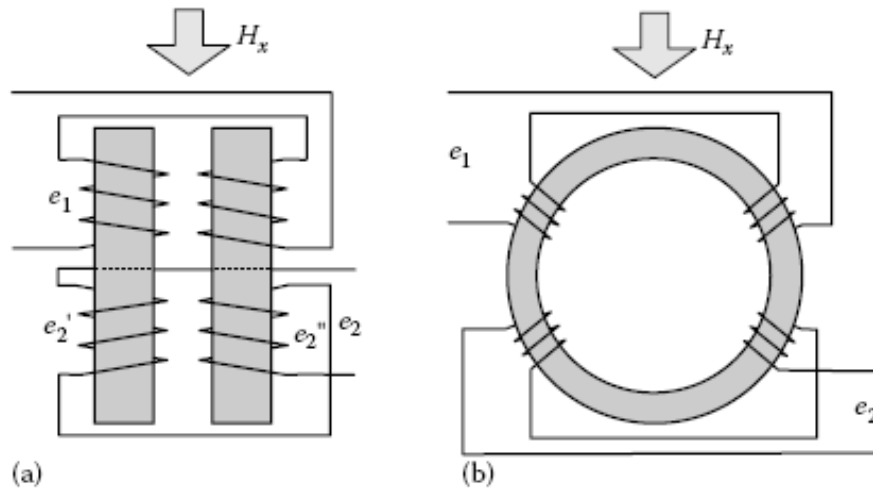


FIGURE 4.51

A fluxgate sensor with two rod cores (a) and ring core fluxgate sensor (b) (bright).

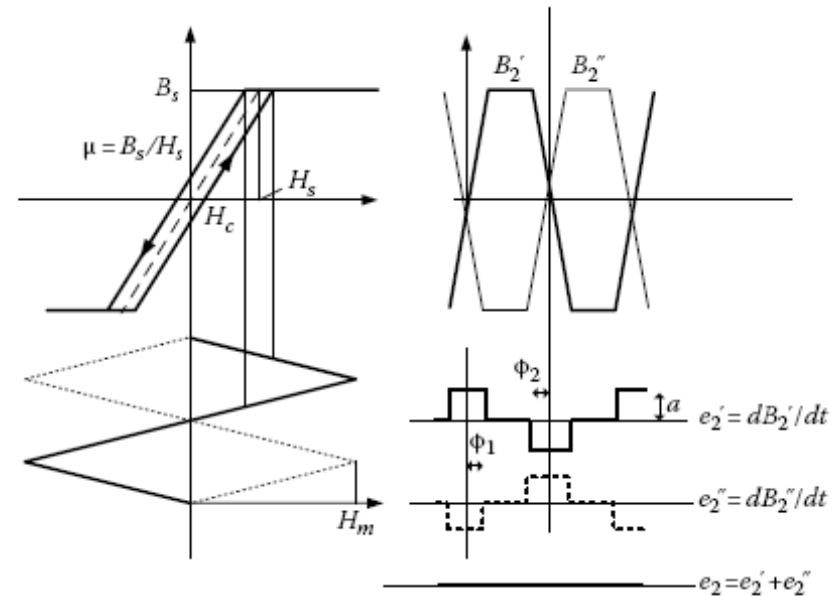


FIGURE 4.52

Signals in fluxgate sensor for $H_x=0$.

Considering the sensor with the core represented by simplified hysteresis shape and the exciting signal by a simplified triangular shape (Figure 4.52). The output signal in secondary coil with n_2 turns wound on the core with area A is:

$$e_2 = -An_2 \frac{dB}{dt} = 0.$$

حسگرهای شارگیت (Fluxgate):

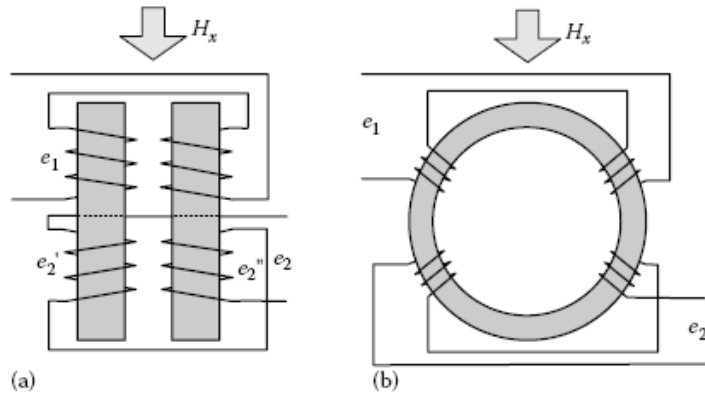


FIGURE 4.51
A fluxgate sensor with two rod cores (a) and ring core fluxgate sensor (b) (bright).

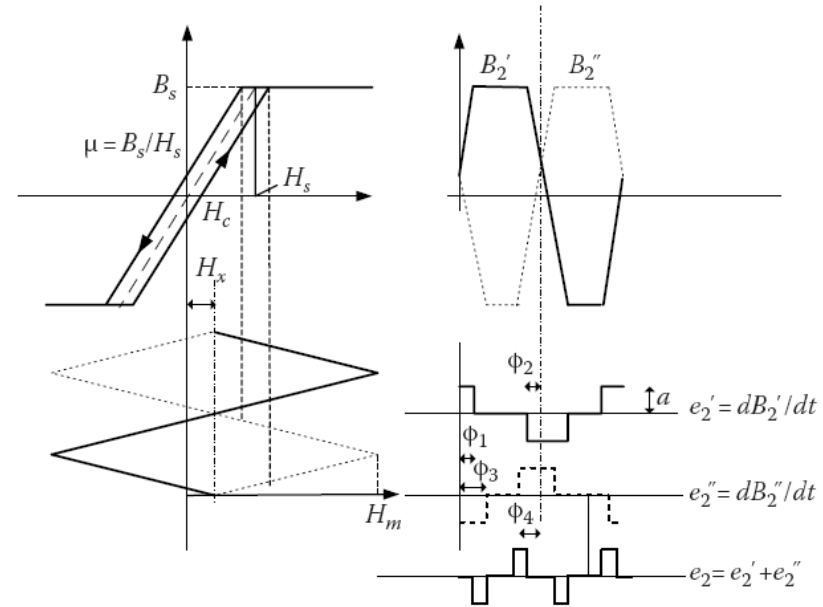


FIGURE 4.53
Signals in a fluxgate sensor for some external field H_x .

$$e_2 = e_2' + e_2'' = 16n_2fA\mu H_x \sin \pi \frac{H_s}{H_m} \sin \left(2\omega t - \pi \frac{H_c}{H_m} \right) + \text{L}$$

Thus, the odd harmonics are eliminated and the output comprises only even harmonics. The magnitude of the second harmonic output signal is:

$$E_2 = 16n_2fA\mu H_x \sin \pi \frac{H_s}{H_m}$$

حسگرهای مقاومت مغناطیسی (Magnetoresistor):

The magnetoresistive (MR) effect (change of resistance caused by magnetic field) exists practically in all metals but it is detectable only at very low temperature and in high magnetic field. However, there are several materials for which the MR effect is sufficiently large at room temperature: ferromagnetic metals, semiconductors, and some minerals as bismuth or lanthanum-based oxides. The MR effect can be large in some artificial multilayer structures like valve or superlattices.

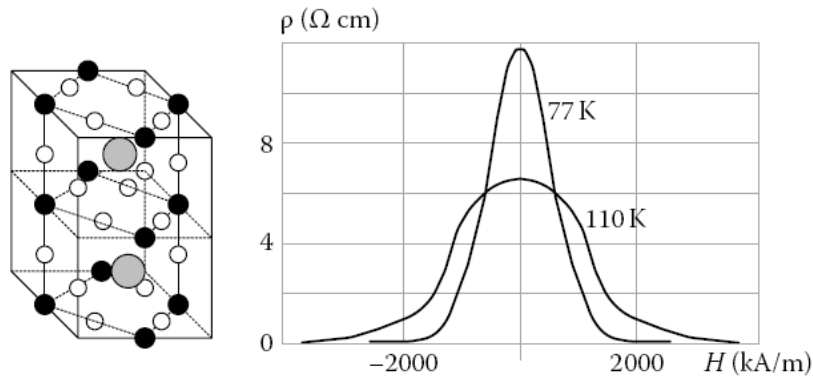


FIGURE 4.74

The crystal structure of manganate perovskites $\text{La}_{0.7}\text{Ca}_{0.3}\text{MnO}_3$ and its change of resistance versus magnetic field. (From McCormack, M. et al., *Appl. Phys. Lett.*, 64, 3045, 1994.)

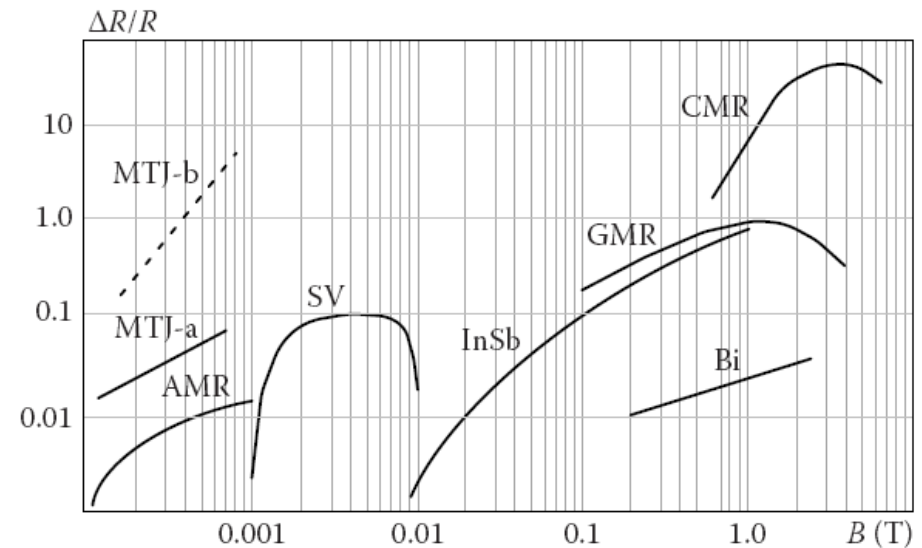


FIGURE 4.73

The main magnetoresistive effects: MTJ, magnetic tunnel junction; AMR, anisotropic magnetoresistance; SV, spin valve; InSb, semiconductor magnetoresistors; GMR, giant multilayer magnetoresistance; Bi, bismuth; CMR, colossal magnetoresistance.

حسگرهای مقاومت مغناطیسی (Magnetoresistor):

In magnetoresistance (MR) effect, a conductor changes its electric resistance in the presence of an external magnetic field. The MR effect was discovered by William Thomson (*Lord Kelvin*) in 1856. He found that iron and nickel exhibited a small increase in electrical resistance along the direction of an applied magnetic field and a similar decrease in resistance in the transverse direction. The magnitude of the effect, MR (unitless), depends on the material of the conductor as:

$$M_R = \frac{\rho_B - \rho_0}{\rho_0} \times 100\%$$

where ρ_B (in $\Omega \cdot m$) is the resistivity of the conductor under an applied magnetic field B , and ρ_0 (in $\Omega \cdot m$) is the resistivity of the conductor without an applied magnetic field B .

EXAMPLE 5.6

Under an applied magnetic field $B = 0.75$ T, an iron (Fe) conductor changes its resistivity from $\rho_0 = 10$ ($\mu\Omega \cdot \text{cm}$) to $\rho_{0.75} = 10.53$ ($\mu\Omega \cdot \text{cm}$); while a nickel (Ni) conductor (with the same size and shape as the iron conductor) changes its resistivity from $\rho_0 = 6.99$ ($\mu\Omega \cdot \text{cm}$) to $\rho_{0.75} = 7.07$ ($\mu\Omega \cdot \text{cm}$). Which material displays a stronger magnetoresistance effect?

SOLUTION

For Fe:

$$M_R = \frac{10.53 - 10}{10} = 0.053 \quad \text{or } 5.3\%$$

For Ni:

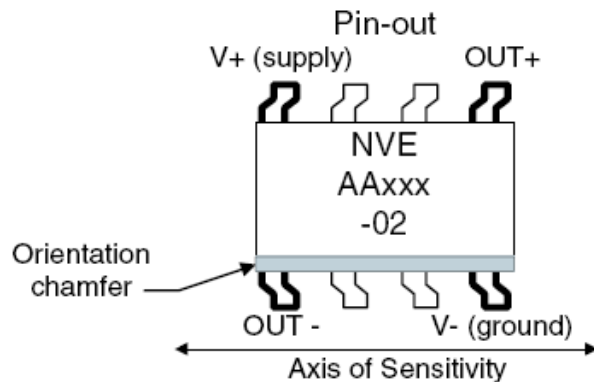
$$M_R = \frac{7.07 - 6.99}{6.99} = 0.011 \quad \text{or } 1.1\%$$

Thus, iron demonstrates a stronger MR effect.

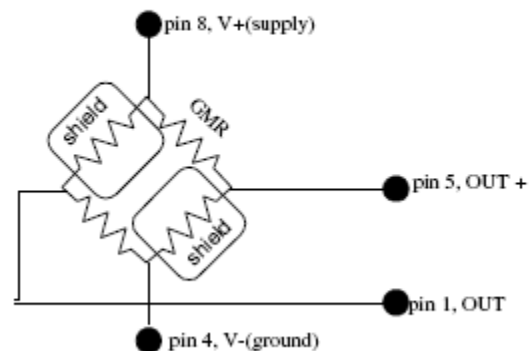
نمونه تجاری مقاومت مغناطیسی GMR:

NVE AAxxx-02 GMR Sensor

NVE's Giant Magnetoresistive Field Sensors offer unique and unparalleled magnetic sensing capabilities. The high sensitivity and ability to sense static magnetic fields provides superior performance which set them apart from other sensors on the market today. NVE's sensors provide high sensitivity, temperature stability, low power consumption, and small size.



Functional Block Diagram



NVE's Sensors can be applied to :

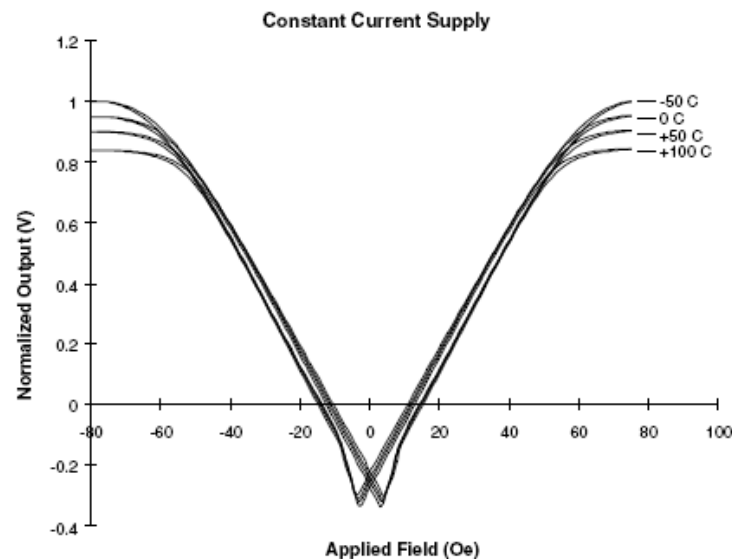
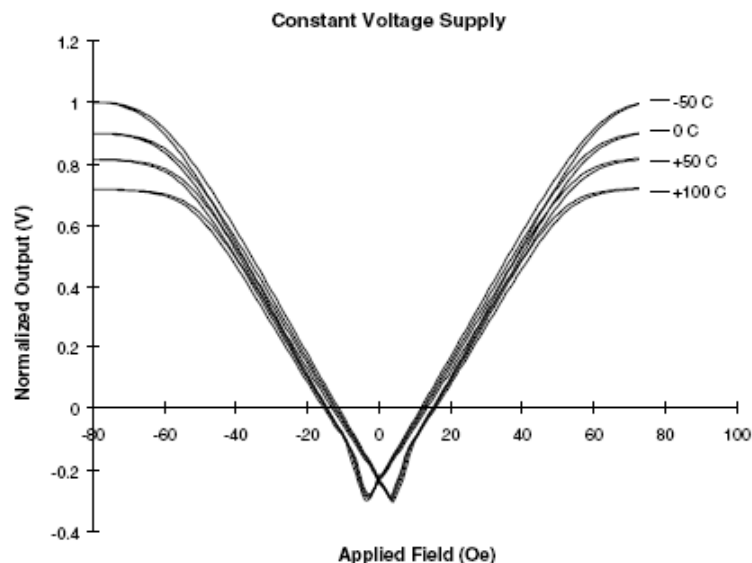
- Proximity Sensing
- Motion, Speed, and Position Sensing
- Current Detection
- Magnetic Media Detection
- Synchronization
- Earth's Field Sensing

نمونه تجاری مقاومت مغناطیسی GMR

NVE AAxxx-02 GMR Sensor

Magnetic Characteristics (5 k Ω $\pm$ 20% bridge)

Part Number	Saturation Field (Oe)	Specified Linear Range (Oe)		Sensitivity (mV/V/Oe)	
		Min	Max	min	max
AA002-02	15	0	10.5	3	4.2
AA003-02	20	0	14	2	3.2
AA004-02	50	0	35	0.9	1.3
AA005-02	100	0	70	0.45	0.65



نمونه تجاری مقاومت مغناطیسی GMR:

NVE AAxxx-02 GMR Sensor

General Characteristics Magnetic Field Sensors

Property	Min	Nominal	Max	Unit
Input Voltage Range			$\pm 25^4$	V
Operating Frequency	DC		$> 1^5$	MHz
Temperature Range	-50		125^4	°C
Electrical Offset (V)	-4		4	mV/V
Max Output		45^1		mV/V
Nonlinearity			2^6	% (unipolar)
Hysteresis			4^6	% (unipolar)
TCR		+0.14		% / K
TCOI		+0.03		% / K
TCOV		-0.1		% / K
Off-axis Characteristic		$\cos. \beta^7$		
ESD		400		V pin to pin HBM



Superconductivity was discovered in 1911 by H. Kamerlingh Onnes in Holland during study of the electrical resistance of frozen mercury as a function of temperature. When cooling Hg to the temperature of liquid helium, he found that the resistance vanished abruptly at approximately 4 K ($= -269.15^{\circ}\text{C}$). Onnes was the first to liquify helium in 1908. In 1913 he won the [Nobel Prize](#) for his liquification of helium and discovery of superconductivity.

ابر رساناهای دما بالا و دما پایین:

TABLE 2.12

Typical Superconducting Materials

LTS Materials		HTS Materials	
Material	T_c (K)	Material	T_c (K)
Al	1.2	YBa ₂ Cu ₃ O ₇	92
Hg	4.2	Bi ₂ Sr ₂ Ca ₂ Cu ₃ O ₁₀	110
Pb	7.2	Tl ₂ Bi ₂ Ca ₂ Cu ₃ O ₁₀	125
Nb	9.3	HgBa ₂ Ca ₂ Cu ₃ O ₈	135
Nb ₃ Al	16.0		

دو خاصیت اصلی ابر رسانا: ۱- مقاومت الکتریکی صفر ($R=0$); ۲- دیامغناطیس کامل ($\mu=0$)

- دمای بحرانی
- شدت میدان مغناطیسی بحرانی
- وابستگی دمایی شدت میدان مغناطیسی بحرانی
- جریان بحرانی

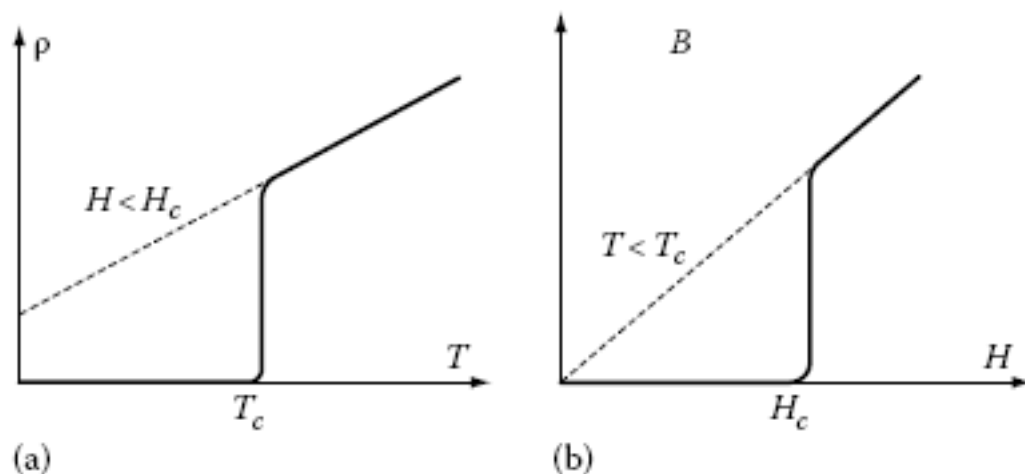
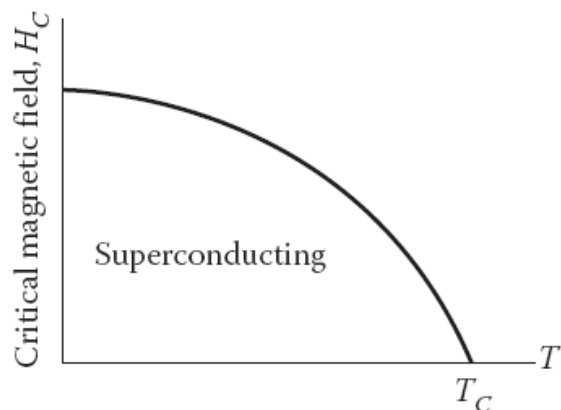


FIGURE 2.60

The material becomes a perfect conductor (a) and perfect diamagnetic (b) at low temperature.

دو خاصیت اصلی ابر رسانا: ۱- مقاومت الکتریکی صفر ($R=0$); ۲- دیامغناطیس کامل ($\mu=0$)



Temperature dependence of the critical field H_C .

- دمای بحرانی
- شدت میدان مغناطیسی بحرانی
- وابستگی دمایی شدت میدان مغناطیسی بحرانی
- جریان بحرانی

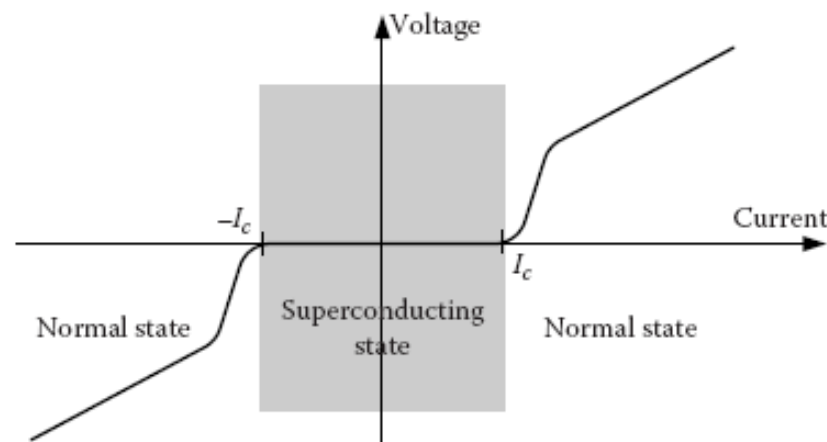


FIGURE 4.159
Voltage versus current of a shunted Josephson junction.

Meissner Effect

ابر رسانی:

The *Meissner–Ochsenfeld effect*, discovered in 1933, states that magnetic fields are expelled from superconducting materials when the materials are cooled below their *critical* or *transition* temperature, T_c . This feature differs from that of normal conducting materials in which magnetic flux lines can penetrate through the material.

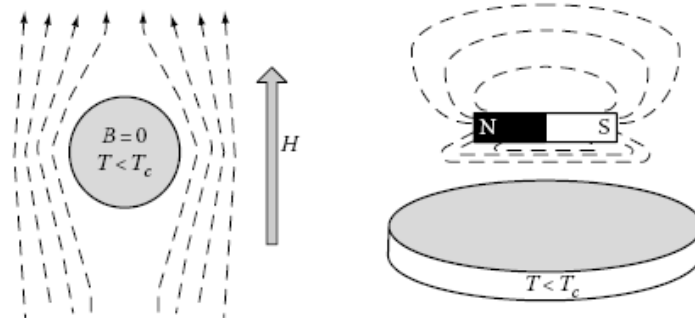
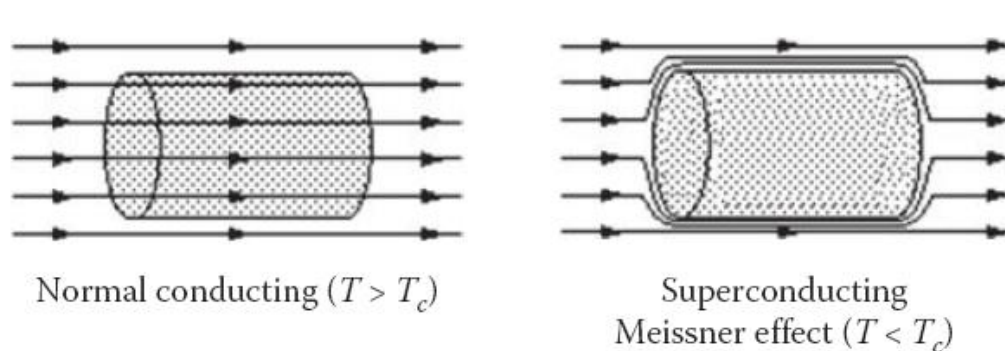


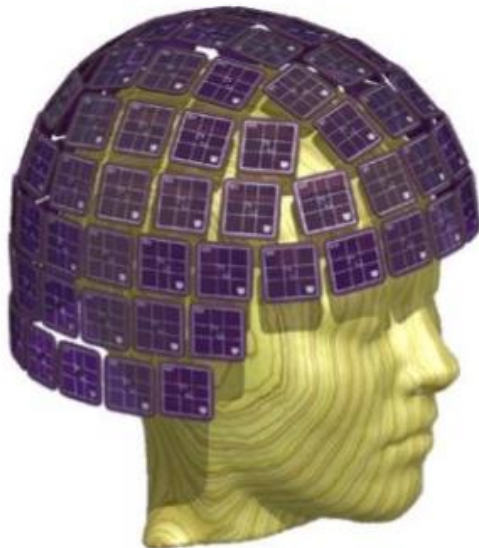
FIGURE 2.61
Two experiments illustrating the Meissner effect.

screening current, i_s

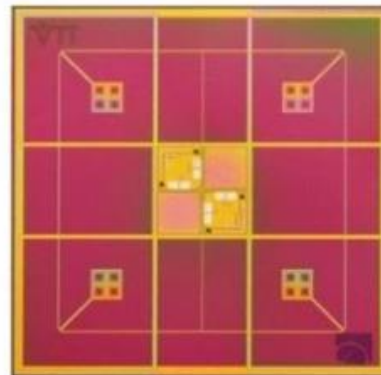
According to the Meissner effect, the current i_s flows only in a very thin boundary area called London penetration depth λ_L —for niobium, this thickness is about 40nm. This current acts as a perfect screen and therefore, the superconducting material is perfectly diamagnetic so that it does not allow the external magnetic flux to penetrate the inside the material.

ابر رساناها به عنوان حسگرهای SQUID برای تشخیص میدانهای مغناطیسی ضعیف کاربرد دارند.
یک کاربرد ابرساناها تصویر نگاری از فعالیت مغز است:

SQUID Sensors



- 306 independent measurement channels.
- Organized in channel-triplets on 102 silicon chips:
 - Two planar gradiometers
 - One radial magnetometer



فیلمهایی در معرفی ابر رسانا:

<https://www.aparat.com/v/yo1bL>

مطالب جالبی از کاربرد و آثار ابر رساناها در youtube موجود است.



تشکر از توجه شما!
سوال؟

